## Decision Analysis – Assignment



Test Content

John has just graduated from the university and he is looking for a job. John has 4 criteria when considering job positions, namely **renumeration package (R), job prospects (P), job location (L) and flexi working hours (F).**

John has come up with the following pairwise comparison matrix of criteria:

| | R | P | L | F |
|---|---|---|---|---|
| R | 1 | 5 | 2 | 3 |
| P | 1/5 | 1 | 1/4 | 1/6 |
| L | 1/2 | 4 | 1 | 1/2 |
| F | 1/3 | 6 | 2 | 1 |

At present, John has 4 job offers **J1, J2, J3, J4.** He has compared the job offers with respect to each criterion and come out with the following comparison matrices:

**Renumeration package (R)**

| | J1 | J2 | J3 | J4 |
|---|---|---|---|---|
| J1 | 1 | 2 | 3 | 3 |
| J2 | 1/2 | 1 | 5 | 1/2 |
| J3 | 1/3 | 1/5 | 1 | 1/4 |
| J4 | 1/3 | 2 | 4 | 1 |

**Job prospects (P)**

| | J1 | J2 | J3 | J4 |
|---|---|---|---|---|
| J1 | 1 | 1/2 | 1/3 | 1/4 |
| J2 | 2 | 1 | 1/3 | 1/2 |
| J3 | 3 | 3 | 1 | 4 |
| J4 | 4 | 2 | 1/4 | 1 |

**Job location (L)**

| | J1 | J2 | J3 | J4 |
|---|---|---|---|---|
| J1 | 1 | 4 | 3 | 1/4 |
| J2 | 1/4 | 1 | 2 | 1/3 |
| J3 | 1/3 | 1/2 | 1 | 4 |
| J4 | 4 | 3 | 1/4 | 1 |

**Flexi working hours (F)**

| | J1 | J2 | J3 | J4 |
|---|---|---|---|---|
| J1 | 1 | 5 | 3 | 1/2 |
| J2 | 1/5 | 1 | 3 | 1/3 |
| J3 | 1/3 | 1/3 | 1 | 1/4 |
| J4 | 2 | 3 | 4 | 1 |

**Important: In all your computation, keep to 6 decimal places accuracy.**

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## Question 1

1 Point

What is John's goal?

- A. Choose a job from the 4 job offers
- B. Choose a job from the 4 job offers that best fits his 4 criteria
- C. Look for a job

## Question 2

1 Point

What are John's criteria when he considers a job position?

- A. Renumeration package
- B. Job location
- C. Job prospects
- D. Flexi working hours
- E. All those mentioned in the choices

## Question 3

1 Point

What are the decision alternatives for John?

- A. J3
- B. J4
- C. J1
- D. J2
- E. All those mentioned in the choices

**Compute the priority of each criterion in terms of its contribution to the overall goal.**

## Question 4

4 Points

Which is the most important criterion?

- A. Renumeration package
- B. Job prospects
- C. Job location
- D. Flexi working hours

## Question 5

4 Points

Which is the second most important criterion?

- A. Renumeration package
- B. Job prospects
- C. Flexi working hours
- D. Job location

## Question 6

4 Points

Which is the third most important criterion?

- A. Renumeration package
- B. Job prospects
- C. Flexi working hours
- D. Job location

## Question 7

4 Points

Which is the least important criterion?

- A. Flexi working hours
- B. Renumeration package
- C. Job prospects
- D. Job location

Write down the priorities of renumeration package (R), job prospects (P), job location (L) and flexi working hours (F). **Give your answers in 3 decimal places, standard rule of rounding up/down applies.**

## Question 8

7 Points

Priority of renumeration package (R) is

Add your answer

Integer, decimal or E notation allowed

## Question 9

7 Points

Priority of job prospects (P) is

Add your answer

Integer, decimal or E notation allowed

## Question 10

7 Points

Priority of job location (L) is

Add your answer

Integer, decimal or E notation allowed

## Question 11

7 Points

Priority of flexi working hours (F) is

Add your answer

Integer, decimal or E notation allowed

## Question 12

15 Points

Compute the consistency ratio (CR). **Give your answer in 3 decimal places, standard rule of rounding up/down applies.**
Note: For 4 criteria, the consistency index of a randomly generated pair-wise comparison matrix is 0.90.

CR=

Add your answer

Integer, decimal or E notation allowed

## Question 13

2 Points

Is the consistency ratio acceptable?

- A. Yes
- B. No

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Calculate the overall priority of each job offer. **Give your answer in 3 decimal places, standard rule of rounding up/down applies.**

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## Question 14

7 Points

Overall priority of J1 is

Add your answer

Integer, decimal or E notation allowed

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## Question 15

7 Points

Overall priority of J2 is

Add your answer

Integer, decimal or E notation allowed

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## Question 16

7 Points

Overall priority of J3 is

Add your answer

Integer, decimal or E notation allowed

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## Question 17

7 Points

Overall priority of J4 is

Add your answer

Integer, decimal or E notation allowed

## Question 18

8 Points

Give your recommendation to John.

- A. John should choose job J3
- B. John should choose job J2
- C. John should choose job J1
- D. John should choose job J4

Questions Filter (18)

Save and Close

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